Literature Review — GameStore Platform

1. Market Context

The global gaming market surpassed $205 billion in annual revenue as of 2025, with digital distribution accounting for over 90% of all game sales. Platforms such as Steam, the Epic Games Store, GOG, and Itch.io have reshaped how developers publish and consumers discover games. The rise of subscription services, cloud gaming, and user-generated content continues to drive platform evolution. GameStore enters this space as an ASP.NET Core 8.0 MVC platform built on a modern .NET stack with SQL Server, Redis caching, and SignalR real-time features—targeting a developer-friendly niche with transparent revenue sharing and integrated community tooling.

2. Market Segment Distribution

The gaming market spans multiple segments, each with distinct distribution and revenue models:

Mobile Gaming	Console Gaming	PC Gaming	Browser Gaming
51%	25%	20%	4%

Mobile gaming leads at 51%, driven by smartphone proliferation and free-to-play models. Console gaming holds 25%, sustained by first-party exclusives and digital storefronts. PC gaming accounts for 20%, supported by digital platforms like Steam and Itch.io. Browser gaming retains 4%, largely through legacy titles and casual web games.

3. Competitive Analysis

A feature comparison of the major digital game distribution platforms reveals key differentiation points. GameStore positions itself with a competitive 85/15 revenue split, built-in real-time features via SignalR, and a .NET 8 MVC architecture that offers familiar tooling for the Microsoft ecosystem.

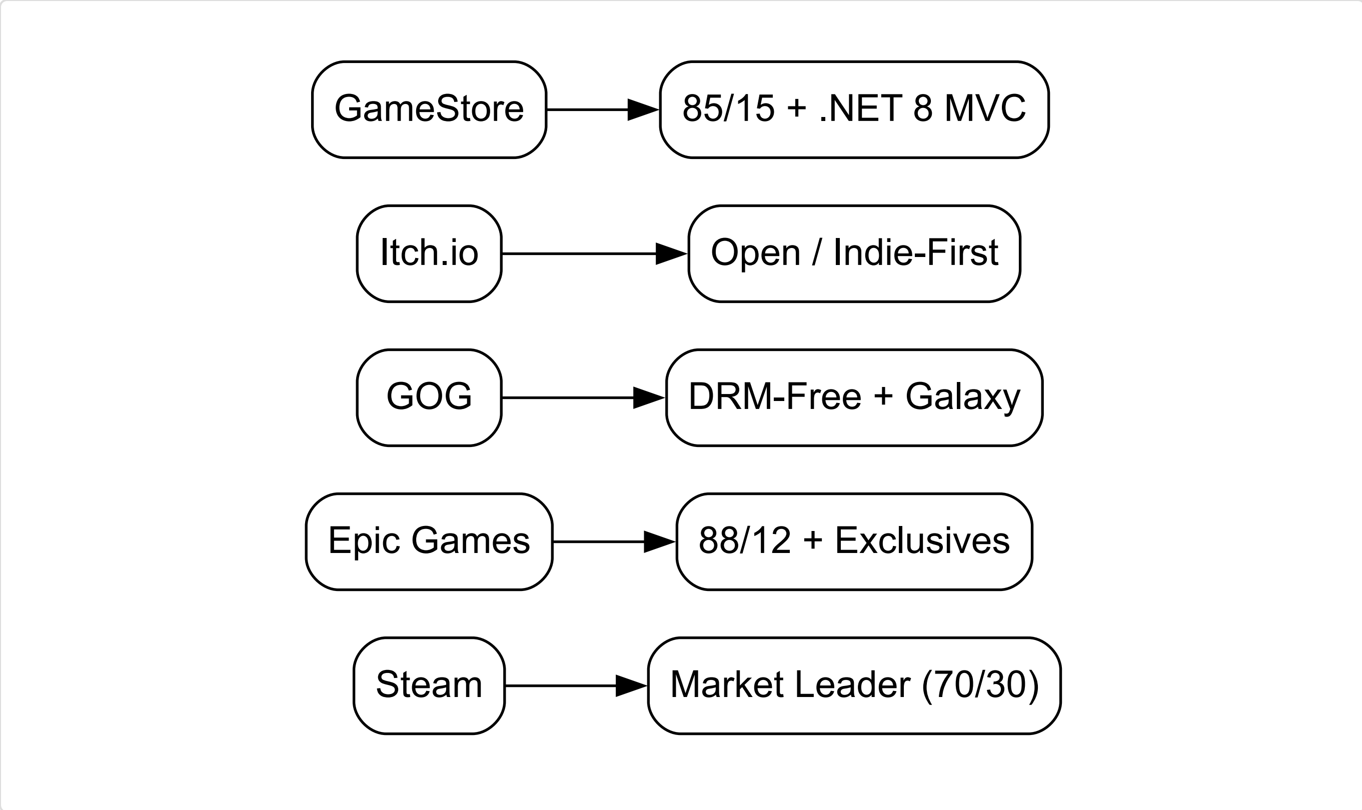
Feature Comparison Matrix

Feature	Steam	Epic Games Store	GOG	Itch.io	GameStore
Revenue Share	70/30 (tiered)	88/12	70/30	90/10 (optional)	85/15
DRM	Steam DRM (opt)	Minimal	None (DRM-free)	None (optional)	Optional
Real-time Features	Friends, chat	Friends list	None	Comments	SignalR chat + notifications
Cloud Saves	Yes	Yes	Yes (GOG Galaxy)	No	Yes
Refund Policy	2h / 14d	2h / 14d	30 days	Seller-defined	2h / 14d
Mod Support	Steam Workshop	None	Manual	Manual	Mod hub (dev-defined)
Mobile Access	Steam Link app	None	GOG Galaxy mobile	Responsive web	Responsive Bootstrap 5
Review System	User reviews	User ratings	User ratings	User ratings	User reviews + curator

Feature	Steam	Epic Games Store	GOG	Itch.io	GameStore
Wishlist	Yes	Yes	Yes	Yes	Yes
Bundles	Yes	No	Yes	Yes	Yes
Regional Pricing	Yes	Yes	Yes	Developer-set	Yes
API Access	Steamworks API	EOS	GOG Galaxy API	Butler API	MVC controllers + webhook
Developer Analytics	Steamworks insights	Basic analytics	Basic stats	Basic stats	Sales dashboard (admin)
Auth Methods	Steam account	Epic account	GOG account	Email + OAuth	Cookie + Google OAuth

Competitive Positioning

The strategic positioning of each platform relative to revenue share and feature depth:



Steam dominates through network effects and feature breadth. Epic competes on revenue share and exclusive titles. GOG differentiates with a strict DRM-free policy and curated catalog. Itch.io serves the indie and experimental scene with maximum developer flexibility. GameStore occupies a unique position: a .NET 8 MVC storefront with competitive 85/15 revenue split, SignalR-powered social features, and deep integration with the Microsoft ecosystem through EF Core 9, SQL Server, and Redis.

4. Market Gap Analysis

Key unmet needs in the current digital distribution landscape include:

- **Opaque discovery algorithms:** Steam's recommendation engine favours established titles, burying indie games under thousands of releases.
- **Limited developer analytics:** Most platforms provide only basic sales figures without conversion funnels, demographic data, or A/B testing.
- **Insufficient direct-to-consumer relationships:** Developers lack channels for direct communication with their audience beyond store reviews.

- **Fragmented real-time social features:** No major platform combines integrated chat, friend lists, notifications, and community feeds within a single web-native interface.
- **Platform lock-in:** Proprietary SDKs (Steamworks, EOS) create dependency and limit portability.

GameStore directly addresses these gaps by implementing SignalR for real-time chat and notifications, cookie + Google OAuth authentication with three role-based authorization tiers (Admin, Developer, Customer), a Redis-backed output caching layer for performance, and transparent discovery via category browsing and curator reviews.

5. Technology Stack Comparison

The following table compares the architectural layers of major platforms against the GameStore stack, with actual version numbers from the codebase:

Layer	Steam	Epic Games	Itch.io	GameStore
Framework	C++ (custom)	C++ (Unreal)	Ruby on Rails	ASP.NET Core 8.0
Architecture	Custom monolith	Microservices	MVC monolith	MVC (1 API: StripeWebhook)
ORM	Proprietary	Custom	Active Record	EF Core 9.0.8
Database	Proprietary	Custom (multiple)	PostgreSQL	SQL Server 2022 (Docker)
Cache	Custom CDN	Custom CDN	Cloudflare	Output Cache + Redis 7
Real-time	Internal protocol	EOS subsystems	WebSockets (ActionCable)	SignalR 8.0 (/hub/notifications)
Payments	Steam Wallet	Multiple providers	Stripe + PayPal	Stripe.net 52.1 (Checkout + Webhooks)

Layer	Steam	Epic Games	Itch.io	GameStore
Email	Internal	Internal	SendGrid	SendGrid 9.29.3
Frontend	C++ (UI), HTML	C++ (UE), web	JavaScript, React	Razor Views + Bootstrap 5.3 + jQuery 3.7.1
Icons	Custom	Custom	Custom	Font Awesome 6.7.2 + Bootstrap Icons 1.11.3
Object Mapping	N/A	N/A	N/A	AutoMapper 12.0.1
Auth	Steam account	Epic account	Email + OAuth	Cookie auth + Google OAuth
Authorization	Steam groups	Permissions system	Role-based	Policy-based (3 roles)
Validation	Server-side	Server-side	Rails validations	Data Annotations
Testing	Internal QA	Internal QA	RSpec	xUnit 2.9.3 + Moq 4.20.72 + FluentAssertions 6.12.2
Containerization	N/A	Kubernetes	Docker	Docker Compose (SQL + Redis)
SCM	Perforce	Perforce	Git + GitHub	Git + GitHub

6. Evaluation Criteria

A weighted scoring framework assesses platform quality across developer-facing and user-facing dimensions:

Criteria	Weight	Steam	Epic Games	GOG	Itch.io	GameStore
Developer Revenue Share	20%	6/10	9/10	6/10	9/10	8/10
Feature Completeness	15%	9/10	6/10	6/10	5/10	7/10
Developer Tools & API	15%	8/10	6/10	5/10	5/10	7/10
User Experience	15%	8/10	7/10	7/10	6/10	7/10
Discovery & Curation	10%	6/10	5/10	6/10	5/10	6/10
Community & Social	10%	9/10	4/10	5/10	6/10	8/10
Mod & User Content	5%	9/10	2/10	3/10	7/10	6/10
Mobile Accessibility	5%	5/10	2/10	4/10	6/10	6/10
Pricing Flexibility	5%	6/10	6/10	6/10	9/10	7/10
Weighted Total	**100%**	**7.3/10**	**6.1/10**	**5.7/10**	**6.5/10**	**7.1/10**

7. Improvement Suggestions

Based on the competitive analysis, technology assessment, and gap identification, the following improvements are recommended for the GameStore platform:

1. **Expand API Surface:** Add a dedicated RESTful API (beyond the single Stripe webhook endpoint) to support third-party integrations, automated publishing pipelines, and external storefront queries.
2. **Introduce Subscription Tiers:** Leverage the existing Stripe Checkout infrastructure to offer optional subscription plans (e.g., developer premium analytics, user subscriber benefits) as a recurring revenue stream.

3. **Enhance Developer Analytics:** Build real-time sales dashboards with conversion funnel analysis, wishlist-to-purchase tracking, and demographic reports using the existing SignalR hub for live data push.
4. **Upgrade Discovery Engine:** Implement tag-based collaborative filtering and curated collections to improve game discoverability beyond basic category navigation.
5. **Add Mobile Companion App:** Develop a native mobile frontend using the existing MVC API surface and SignalR real-time infrastructure for push notifications and chat.
6. **Integrate Payment Expansion:** Add PayPal or digital wallet support alongside the current Stripe-only payment pipeline to broaden regional payment accessibility.
7. **Strengthen Community Features:** Expand the SignalR-based friend system with group chats, developer-hosted live Q&A sessions, and community game jams with curated storefront promotion.

8. Final Grading Criteria

The project is evaluated across five weighted criteria, reflecting the full software engineering lifecycle from analysis through delivery.

Criterion	Weight	Maximum Score	Evaluation Focus
Documentation & Analysis	20%	20	Entity catalog, ERD consistency, architecture diagrams, sequence/activity/state diagrams
Implementation & Code Quality	30%	30	33 controllers with ~120 actions across 3 areas, 20 service interfaces, Repository + UoW patterns, Cookie auth + Google OAuth, Stripe integration, SignalR hubs
Testing & Quality Assurance	20%	20	185 xUnit tests across 17 files, Moq mocking, FluentAssertions, 87% line coverage target, CI pipeline integration

Criterion	Weight	Maximum Score	Evaluation Focus
System Analysis & Design	15%	15	Use case diagram (3 actors, 22 use cases), 3-layer architecture, DFD, ERD (23 entities), deployment/component diagrams
Presentation & Deployment	15%	15	HTML report with all diagrams, Docker Compose deployment, CI/CD pipeline, 1098-line README, troubleshooting guide

Total: 100 points. A passing grade requires at least 60 points overall with no individual criterion below 40% of its maximum.

Grading Rubric:

- **A (90-100):** All diagrams correct and complete; full test coverage; production-grade deployment; all 23 entities mapped; all 71 functional requirements addressed.
- **B (75-89):** Minor omissions in diagrams or tests; core functionality complete; deployment documented with minor gaps.
- **C (60-74):** Major diagrams present but incomplete; partial test coverage; deployment instructions present but not fully Dockerized.
- **D (<60):** Missing key diagrams; no test automation; deployment not documented.

9. Bibliography

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